Maclaurin Series & Convergence of Series

Section 1: The Maclaurin Series Formula

1 What is the general formula for the Maclaurin series expansion of a function $f(x)$?

a. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$

b. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n$

c. $f(x) = \sum_{n=1}^{\infty} \frac{f(n)}{n!}x^n$

d. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!}x^n$

2 Find the first three non-zero terms of the Maclaurin expansion of $f(x) = \cos(2x)$.

a. $1 - 2x^2 + \frac{2}{3}x^4$

b. $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

c. $1 - 4x^2 + 16x^4$

d. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

3 What is the coefficient of x^3 in the Maclaurin series of e^{-3x}?

a. $-\frac{9}{2}$

b. $\frac{9}{2}$

c. -9

d. -27

Section 2: Manipulating Standard Maclaurin Expansions

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4 Find the Maclaurin series for $\ln(1+3x)$ up to the x^3 term.

a. $3x - \frac{9}{2}x^2 + 9x^3$

b. $3x + \frac{9}{2}x^2 + 9x^3$

c. $x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

d. $3x - 9x^2 + 27x^3$

5 Evaluate $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$ using Maclaurin series.

a. $\frac{1}{2}$

b. 1

c. 0

d. ∞

6 Find the first two non-zero terms of the series for $\frac{1}{1+x^2}$.

a. $1 - x^2$

b. $1 + x^2$

c. $1 - x$

d. $x - x^3$

Section 3: The Ratio Test for Convergence

7 What does the Ratio Test state for a positive series $\sum a_n$ when $L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$?

a. Converges if $L < 1$, diverges if $L > 1$, test is inconclusive if $L = 1$

b. Converges if $L = 1$, diverges otherwise

c. Diverges if $L < 1$, converges if $L > 1$

d. Always inconclusive for power series

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8 Determine the convergence of $\sum_{n=1}^{\infty} \frac{n}{3^n}$ using the Ratio Test.

a. Converges, because $L = \lim \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{1}{3} < 1$

b. Diverges, because $L = 3 > 1$

c. Inconclusive, because $L = 1$

d. Diverges, because the terms do not approach 0

9 What is the limit L in the Ratio Test for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. 1 (test is inconclusive)

b. 0

c. ∞

d. $1/2$

Section 4: Radius and Interval of Convergence

10 Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$.

a. $R = \frac{1}{2}$

b. $R = 2$

c. $R = 1$

d. $R = \infty$

11 What is the radius of convergence of the Maclaurin series for e^x?

a. $R = \infty$

b. $R = 1$

c. $R = e$

d. $R = 0$

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12 For $\sum_{n=0}^{\infty} x^n$, what is the exact interval of convergence?

a. $(-1, 1)$

b. $[-1, 1]$

c. $(-1, 1]$

d. $[-1, 1)$

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